LEVEL 01

Adaptive Clinical Recruitment

01

What This Benchmark Is

Long-horizon trial planning, reduced to an interface you can test and a score you can inspect.

CORE LOOP

A benchmark for sequential trial planning.

One run carries an agent through screening, follow-up, site allocation, and retention pressure over 180 simulated days.

simulated days

public tasks

implemented actions

WHAT THE AGENT SEES

Typed observations plus a final grade.

Action-specific candidate pools, site metrics, milestone state, plan state, memory state, token signals, and counterfactual hints stay visible during the episode.

Episodes end with final_score in (0, 1), so step-level reward and end-of-run grading can be inspected together.

A single run advances through candidate screening, consent pressure, site allocation, and retention risk before the grader emits final_score in (0, 1).

Core truth: 180 steps / 3 tasks / 8 actions / 37 training features.

Interpretation: workflow benchmark first, leaderboard second.

Source: corrected paper, regenerated charts, and fresh 5-seed sweep.

02

How The Environment Moves

A patient funnel with delayed consequence, typed state, and action-specific routing.

CANDIDATE ROUTING

Each action pulls from its own pool.

screen_patient uses available_patients.

recontact uses recontact_candidates.

allocate_to_site uses allocation_candidates plus a site_id.

LONG-HORIZON SIGNALS

The environment keeps delayed state in view.

milestones, active_constraints, delayed_effects_pending, uncertainty_components, and token_efficiency_score all remain observable.

PLANNING AND MEMORY

The benchmark exposes internal coordination surfaces.

current_plan, indexed_memory_summary, retrieved_memory_context, active_milestone, and counterfactual_hint let agents act beyond one-step reflexes.

PHASE LADDER

- new candidates appear in available_patients
- follow-up pressure surfaces in recontact_candidates
- consented patients move into allocation_candidates
- dropout risk remains active after enrollment
- constraints and bottlenecks alter what should happen

Level 02 / mechanics: candidate pools, delayed effects, milestones, and explicit recovery.

Quiet reference: the corrected evaluation path matters because interface truth changes what results mean.

Read with: models.py, env.py, training/neural_policy.py.

What Is Actually Implemented

The corrected docs, code path, and baseline surface after the benchmark audit.

03

CORE FILES

The main benchmark path is compact enough to audit.

models.py defines the typed interface.

env.py implements the dynamics and reward path.

training/neural_policy.py exports ACTION_SPACE and STATE_DIM = 37.

REPO BASELINES

Four trainable baselines share one backbone.

HCAPO, MiRA, KLong, and MemexRL all sit on the same pure-NumPy actor-critic infrastructure.

SCOPE DISCIPLINE

The corrected docs separate core benchmark claims from auxiliary scaffolds.

The paper and public docs now avoid treating every research helper as a validated contribution.

IMPLEMENTED INTERFACE

SCREEN_PATIENT

RECONTACT

8 ACTIONS

ALLOCATE_TO_SITE

ADJUST_STRATEGY

PLAN_NEXT_PHASE

SUMMARIZE_AND_INDEX

The corrected docs and paper use the exact action surface exported by models.py and training/neural_policy.py.

RETRIEVE_HISTORY

STOP_RECRUITMENT

Implementation truth: 8 actions and 37 features are exported by code, not by marketing copy.

Repo baselines are presented as repo baselines, not as strict external replications.

Read with: full_sweep.py, reproducibility.py, research/methods/.

04

What The Fresh Sweep Says

The corrected 5-seed run narrows the story: active benchmark, clustered baselines, no clear statistical winner.

MEAN SCORE / 95% CI

HCAPO 0.2215

[0.2100, 0.2303]

KLong 0.2152

[0.1977, 0.2286]

MemexRL 0.2148

[0.1943, 0.2352]

MiRA 0.2094

[0.2023, 0.2165]

PAIRWISE TESTS

HCAPO vs MiRA
p = 0.1823

HCAPO vs KLong
p = 0.3849

HCAPO vs MemexRL
p = 0.6370

MiRA vs KLong
p = 0.6674

MiRA vs MemexRL
p = 0.6656

KLong vs MemexRL
p = 0.9756

VERDICT

No pairwise comparison reaches $p < 0.05$.

The corrected benchmark now supports a conservative read: active, reproducible, and not yet clearly separated by the current baseline suite.

Fresh sweep: 5 seeds / 30 training episodes per seed / 15 evaluation episodes per seed.

Result: HCAPO is highest mean at 0.2215, but confidence intervals overlap and no pairwise test reaches $p < 0.05$.

Use this as a stress test narrative, not as proof of decisive baseline dominance.

How To Read It Critically

Treat the repository as a reproducible evaluation bundle first, and a leaderboard only with caution.

05

STRONGEST CLAIM

Typed, corrected, and reproducible.

The environment path, docs, diagrams, sweep outputs, and paper now tell the same conservative story.

DO NOT OVERREAD

This is not a finished leaderboard.

The current means are close, variance remains non-trivial, and stronger training budgets are still needed.

SCOPE BOUNDARY

Core path first.

Center claims on env.py, models.py, the baselines, full_sweep.py, reproducibility.py, tests, and generated artifacts.

NEXT WORK

Sharper separation, not louder language.

More seeds, larger budgets, and ablations over state channels or action families should come next.

VERIFICATION

- integration checks
- test_env.py
- test_agents.py
- test_research_modules.py
- test_local_serving.py
- anonymous paper build

LOOK HERE FIRST

- README.md
- data/sweep_results/neurips_report.md
- paper/main.pdf
- docs/images/

Strongest claim: typed, corrected, regenerated, and reproducible.

Weakest claim: decisive empirical separation among the current four baselines.

Next work: more seeds, larger training budgets, and sharper ablations over benchmark surfaces.